University College London

Survival Analysis: Parametric Models

Lecture I: Parametric survival analysis in the overall survival framework

F. Javier Rubio

Department of Statistical Science, University College London

Lectures I and II

1. Parametric survival analysis in the **overall survival** framework.
2. Parametric survival analysis in the **relative survival** framework.

Lecture aims

1. Describe the aims of survival analysis (overall survival).
2. Describe parametric approaches to estimating the survival function.
3. Describe two regression models: proportional hazards (PH) and accelerated failure time (AFT).
4. Present software tools for survival analysis, with a real data example.
5. Discuss other parametric, semiparametric and nonparametric alternatives.

Survival analysis

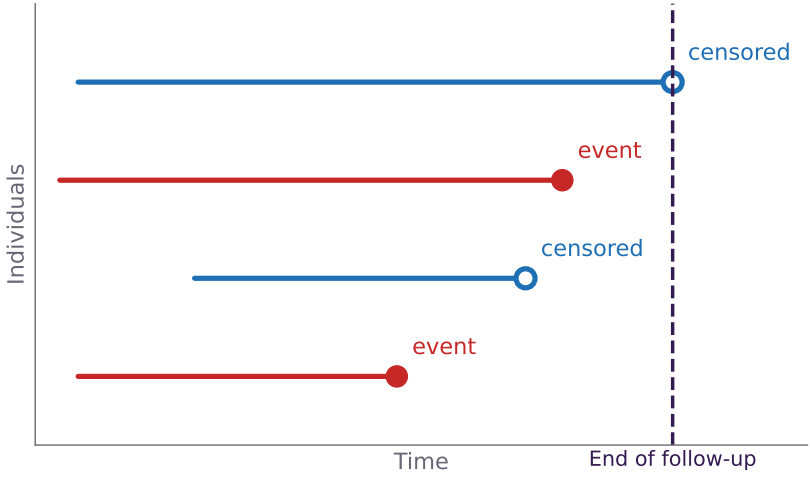
Survival analysis in practice

- In many areas, such as medicine, biology and engineering (reliability), scientists have access to the survival times of a group of individuals or items.
- **Example 1.** The survival of cancer patients after a diagnosis of cancer.
- **Example 2.** The lifespan of a product or device.
- These two scenarios are identical from a theoretical perspective. In practice, however, the first one raises ethical considerations.

The typical data set

- A sample of **times to event**, possibly right-censored, $(t_1, \dots, t_n)$, from a group of individuals.
- The vital status (or **censoring indicators**) $(\delta_1, \dots, \delta_n)$: $\delta_i = 1$ for death, $\delta_i = 0$ for right-censored (alive). Censoring may be due to random drop-out, loss to follow-up, or administrative censoring.
- In some cases, we also know some characteristics of the individuals: **covariates** $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$, such as age, sex, deprivation level, comorbidities or tumour stage.
- **Overall survival:** in this framework, we know the times to event, but we do not consider the cause of death.

Censoring


censored
event
censored
event
Individuals
Time
End of follow-up

Aims of survival analysis

One of the aims of survival analysis is to quantify the survival of a group of individuals. The typical quantities of interest are:

1. The **survival function**

$$S(t) = 1 - F(t) = \mathbb{P}(T > t),$$

   where T is a positive random variable representing the survival time, with cdf F.
2. The **hazard function**, defined as the instantaneous risk:

$$h(t) = \lim_{\Delta t \to 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \overset{\text{(exercise)}}{=} \frac{f(t)}{S(t)}.$$

Survival and hazard functions

- The hazard function and the survival function are linked through the relationship

$$S(t) \overset{\text{(exercise)}}{=} \exp\left\{-\int_0^t h(s)\,ds\right\}.$$

- The function $H(t) = \int_0^t h(s)\,ds$ is known as the **cumulative hazard function**.

The importance of the hazard function

- We are exposed to many forces of mortality: ageing, illnesses, natural disasters, accidents, crime, and so on.
- In biostatistics (and epidemiology), the hazard plays a key role, as it reflects these forces of mortality.

Parametric estimation without covariates

Maximum likelihood estimation: no covariates

- Suppose that $(t_1, \ldots, t_n)$ are independent and identically distributed.
- Suppose that we are interested in estimating the survival function using a parametric distribution $F(\cdot \mid \boldsymbol{\theta})$.
- Which distribution?

A brief catalogue of parametric distributions

- Gamma
- Weibull
- Lognormal
- Log-logistic
- Generalised gamma
- Among many others: see this overview.

Hazard shapes

Different distributions can capture different shapes of the hazard function:

- **Weibull:** increasing, decreasing or constant.
- **Lognormal:** unimodal (up, then down).
- **Generalised gamma:** increasing, decreasing, bathtub or unimodal.

Warning

By selecting a parametric distribution from the catalogue, we are making assumptions about the possible shapes of the true hazard function. Selecting the model using formal tools is usually recommended.

Maximum likelihood estimation: no covariates

- We need to account for the different contributions of the observed and right-censored times:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} f(t_i \mid \boldsymbol{\theta})^{\delta_i} \, S(t_i \mid \boldsymbol{\theta})^{1-\delta_i}.$$

- The MLE $\hat{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ is the value that maximises the likelihood function (numerical methods are usually needed).
- The fitted survival and hazard functions are $S(t \mid \hat{\boldsymbol{\theta}})$ and $h(t \mid \hat{\boldsymbol{\theta}})$.

Software

- Since these are standard methods and distributions, many have already been implemented in R.
- For example, a number of distributions can be fitted to survival data using the flexsurv R package.

Regression models

Regression models: using covariates

- Spoiler: there is no unique way of including covariates in a model for survival times.
- Biostatisticians and epidemiologists tend to think in terms of the hazard, given the interpretation above. Unsurprisingly, most approaches to including covariates consist of **hazard-based regression models**.
- The most popular are the **proportional hazards (PH)** model and the **accelerated failure time (AFT)** model.

Proportional hazards models

- The PH model postulates that the covariates act on a "baseline hazard", either increasing or decreasing it. That is,

$$h_{\mathsf{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) = h_0(t \mid \boldsymbol{\theta}) \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\}.$$

- The corresponding cumulative hazard function is

$$H_{\mathsf{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) \overset{\text{(exercise)}}{=} H_0(t \mid \boldsymbol{\theta}) \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\}.$$

Proportional hazards: interpretation

- $h_0(t \mid \boldsymbol{\theta})$ is the hazard associated with $\mathbf{x}_i = \mathbf{0}$, a sort of "reference point". It can be the hazard of any of the distributions discussed previously.
- If $\mathbf{x}_i^\top \boldsymbol{\beta} > 0$, then $h_{\text{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) > h_0(t \mid \boldsymbol{\theta})$: for a fixed value of $\boldsymbol{\beta}$, the individual's combination of characteristics leads to an **increase** in the hazard compared with the baseline hazard.
- If $\mathbf{x}_i^\top \boldsymbol{\beta} < 0$, then $h_{\text{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) < h_0(t \mid \boldsymbol{\theta})$: the individual's combination of characteristics leads to a **decrease** in the hazard compared with the baseline hazard.

Proportional hazards: hazard ratios

- Let $\mathbf{x}_i$ denote the characteristics of the ith patient and, for a specific $k \in \{1, \ldots, p\}$, let $\tilde{\mathbf{x}}_i$ be equal to $\mathbf{x}_i$ except for its kth entry, $\tilde{x}_{ik} = x_{ik} + 1$.
- Then,

$$\frac{h_{\mathsf{PH}}(t \mid \tilde{\mathbf{x}}_i, \boldsymbol{\theta}, \boldsymbol{\beta})}{h_{\mathsf{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta})} \overset{\text{(exercise)}}{=} \exp\{\beta_k\}.$$

That is, $\exp\{\beta_k\}$ is the **hazard ratio** associated with a one-unit increase in the covariate x_{ik}, with all other covariates held fixed.

Accelerated failure time model

- The AFT model postulates that the covariates act simultaneously on the time scale and on the hazard scale:

$$h_{\mathsf{AFT}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) = h_0\left(t \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right) \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\}.$$

- The corresponding cumulative hazard function is

$$H_{\mathsf{AFT}}(t \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) \overset{\text{(exercise)}}{=} H_0\left(t \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right).$$

AFT model: interpretation

The interpretation of the AFT model is easier if we transform the model. The survival function is

$$
\begin{aligned}
S_{\text{AFT}}(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) &\overset{\text{(exercise)}}{=} \exp\left\{-H_0\left(t_i \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right)\right\} \\
&= S_0\left(t_i \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right) = 1 - F_0\left(t_i \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right) \\
&= \mathbb{P}\left(T_i > t_i \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\} \mid \boldsymbol{\theta}\right) = \mathbb{P}\left(\log T_i > \log(t_i) + \mathbf{x}_i^\top \boldsymbol{\beta} \mid \boldsymbol{\theta}\right) \\
&= \mathbb{P}\left(\log T_i > \log(t_i) - \mathbf{x}_i^\top \boldsymbol{\alpha} \mid \boldsymbol{\theta}\right),
\end{aligned}
$$

where $\boldsymbol{\alpha} = -\boldsymbol{\beta}$.

AFT model: interpretation (continued)

- Let $Y_i = \log(T_i)$ and $Y_i - \mathbf{x}_i^\top \boldsymbol{\alpha} \sim G_0(\cdot \mid \boldsymbol{\theta})$, with pdf g_0.
- Then, writing $y_i = \log(t_i)$,

$$S_{\mathsf{AFT}}(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) \leftrightarrow 1 - G_0\left(y_i - \mathbf{x}_i^\top \boldsymbol{\alpha} \mid \boldsymbol{\theta}\right),$$
$$f_{\mathsf{AFT}}(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta}) \leftrightarrow g_0\left(y_i - \mathbf{x}_i^\top \boldsymbol{\alpha} \mid \boldsymbol{\theta}\right).$$

- We can identify this pdf with that of a **log-linear model**:

$$y_i = \log(t_i) = \mathbf{x}_i^\top \boldsymbol{\alpha} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} G_0(\cdot \mid \boldsymbol{\theta}), \quad i = 1, \ldots, n.$$

- Consequently, the covariates have a direct effect on the log survival time.

Regression models: the likelihood function

- The likelihood function for the PH and AFT models is, for $j \in \{\text{PH}, \text{AFT}\}$,

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \prod_{i=1}^{n} f_j(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta})^{\delta_i} \, S_j(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta})^{1-\delta_i}$$

$$\overset{\text{(exercise)}}{=} \prod_{i=1}^{n} h_j(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta})^{\delta_i} \exp\left\{-H_j(t_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \boldsymbol{\beta})\right\}.$$

- This shows that the likelihood can be written entirely in terms of the hazard function.

Example: lung cancer data

The data

- Survival times t_i, $i = 1, \ldots, 228$, of patients with advanced lung cancer from the North Central Cancer Treatment Group (available as `lung` in the R package **survival**).
- 165 patients died within the follow-up period (1,022 days).

No covariates

- We compare the survival functions of three parametric models: Weibull, lognormal and generalised gamma.
- R code: Example: no covariates

Regression models

- We consider two covariates, age and sex.
- We first compare the PH and AFT models with a lognormal baseline hazard, and then with a Weibull baseline hazard (for which the PH and AFT models coincide).
- R code: Example: regression

Discussion

Discussion

- Parametric survival models are useful tools in many applied areas.
- Their use is not automatic: the user needs to select the parametric model(s).
- PH and AFT models are the most popular hazard-based regression models.
- Confidence intervals for the parameters, and confidence bands for the estimated survival function, can also be obtained.

Extensions: parametric approaches

- The General Hazard (GH) model is an extension of the PH and AFT models.
- The GH model is discussed in Lecture II.
- The GH model is implemented in the R package HazReg.
- Other model structures include the proportional odds model and related extensions.

Extensions: semiparametric and nonparametric approaches

- The **Kaplan–Meier estimator** is a nonparametric estimator of the survival function.
- The **Nelson–Aalen estimator** is a nonparametric estimator of the cumulative hazard function.
- The **Cox model** is a semiparametric version of the PH model.

R code: Cox proportional hazards model

The Cox PH model and the partial likelihood

- The PH model:

$$h_{\mathsf{PH}}(t \mid \mathbf{x}_i, \boldsymbol{\beta}) = h_0(t) \exp\left\{\mathbf{x}_i^\top \boldsymbol{\beta}\right\}.$$

- To avoid misspecifying the baseline hazard, it is often preferred to leave it unspecified (estimating it nonparametrically), while the coefficients $\boldsymbol{\beta}$ are estimated using the log partial likelihood function (Cox 1972):

$$\ell_p(\boldsymbol{\beta}) = \sum_{i:\,\delta_i=1} \mathbf{x}_i^\top \boldsymbol{\beta} - \sum_{i:\,\delta_i=1} \log\left(\sum_{k \in \mathcal{R}(t_i)} \exp\left\{\mathbf{x}_k^\top \boldsymbol{\beta}\right\}\right),$$

where $\mathcal{R}(t) = \{k : t_k \geq t\}$ denotes the risk set at time t.

References

Cox, D. R. 1972. "Regression Models and Life-Tables." *Journal of the Royal Statistical Society. Series B* 34 (2): 187–220.

Harrell, F. E. 2015. *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis*. Springer.

Kalbfleisch, J. D., and R. L. Prentice. 2011. *The Statistical Analysis of Failure Time Data*. Vol. 360. John Wiley & Sons.

Klein, J. P., H. C. Van-Houwelingen, J. G. Ibrahim, and T. H. Scheike. 2016. *Handbook of Survival Analysis*. CRC Press.